Retail Demand Forecasting & Inventory Optimization

A supply chain analytics case study using Python, forecasting and inventory policy

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Executive summary

This project analyzes two years of real online-retail transactions to forecast weekly product demand and convert demand variability into safety-stock and reorder-point recommendations. The workflow demonstrates data preparation, exploratory analysis, model validation, inventory-policy design and executive dashboarding.

1,036,960	GBP 20.1M	4,893	22,951
Valid sales rows	Clean revenue	Products analyzed	Returns / cancellations

Business value

The analysis identifies seasonal demand, compares forecasting methods on unseen weeks, and provides differentiated inventory buffers for high-value products. It shows how transactional data can support replenishment decisions even when inventory-system data are incomplete.

Key result: demand accelerates from September and peaks in November. Inventory parameters should therefore be reviewed before the seasonal build-up rather than after it occurs.

1. Data and analytical approach

Dataset

The UCI Online Retail II dataset contains 1,067,371 transactions from a UK-based online retailer between December 2009 and December 2011. Fields include invoice number, product code, description, quantity, invoice date, unit price, customer identifier and country.

Data preparation

Step	Action
1	Combined the two annual workbook sheets and standardized field names.
2	Separated cancellations and negative-quantity returns from merchandise sales.
3	Retained positive-quantity, positive-price product transactions.
4	Excluded postage, fees, adjustments and other non-merchandise codes.
5	Created weekly and monthly product-demand tables; excluded the final partial week from training.

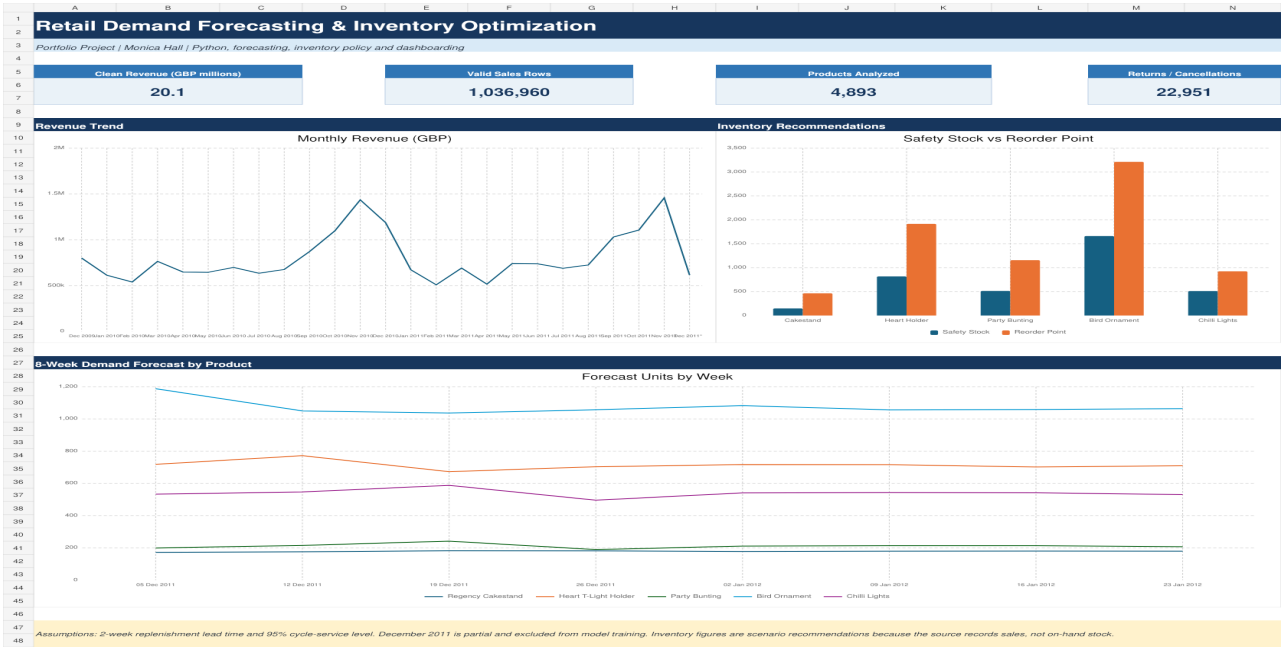
Forecast validation

Five high-value products with at least 90 active weeks and 500 orders were selected. Four-week moving average, seasonal naive (52-week lag), and random forest models were evaluated using the final 13 complete weeks. Mean absolute error was the primary selection metric. The four-week moving average achieved the lowest validation error for all five products.

This is analytically meaningful: a simpler model should be preferred when it performs better on unseen data and is easier for planners to explain and maintain.

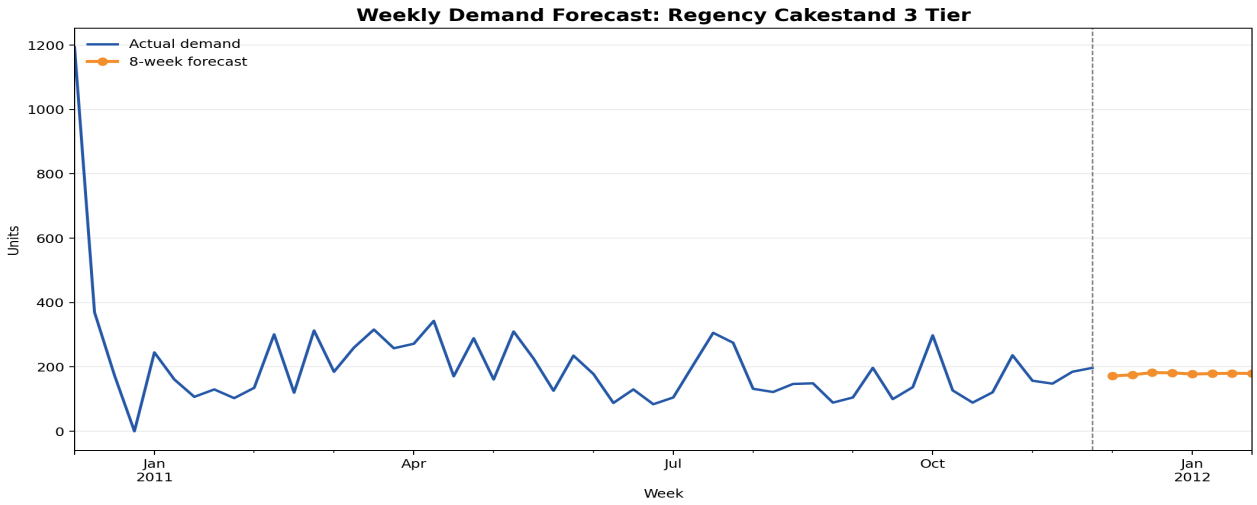
Source: https://archive.ics.uci.edu/ml/datasets/online%2Bretail%2BII

2. Dashboard and demand forecast



The dashboard brings the revenue trend, inventory recommendations and eight-week product forecasts into a single executive view. Revenue rises sharply during September-November in both complete years, demonstrating a repeatable seasonal peak.

Eight-week Regency Cakestand forecast



The selected moving-average model forecasts approximately 1,428 Regency Cakestand units across the next eight weeks. Weekly refreshes are recommended because short-term averages can react quickly to changing demand but may lag abrupt promotional changes.

3. Inventory-policy recommendations

Scenario assumptions

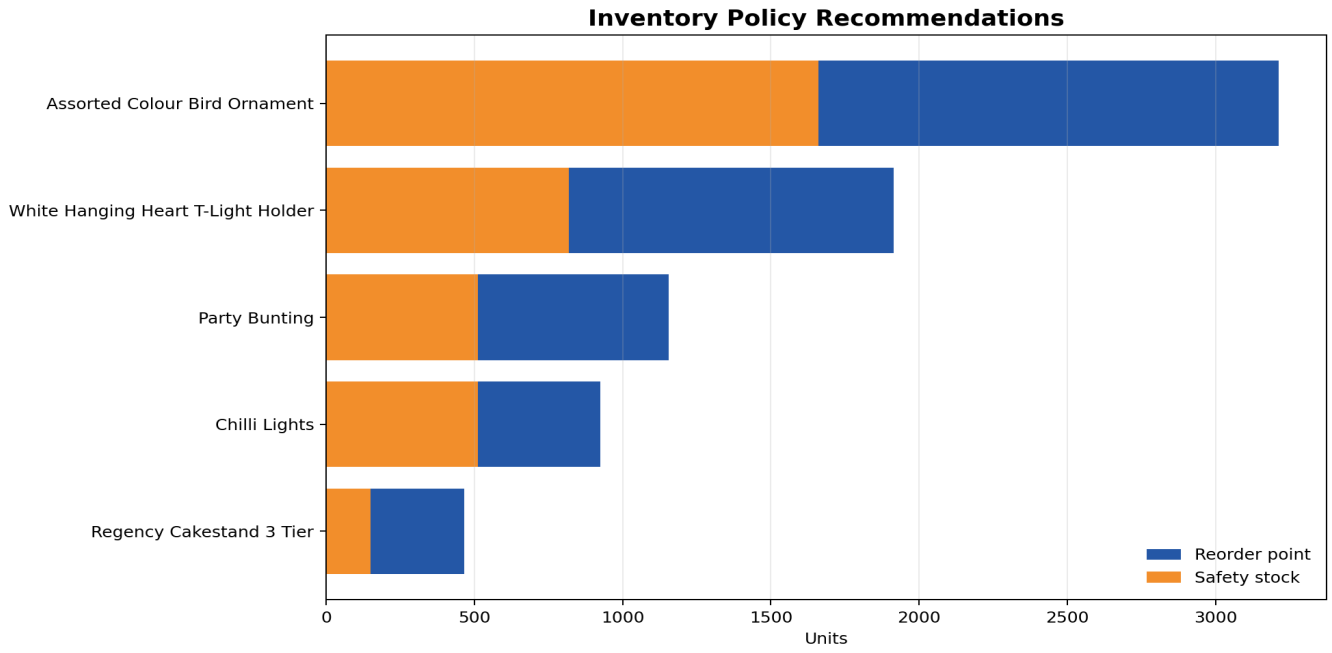
The dataset contains sales transactions but not on-hand stock, supplier lead times or purchase orders. The analysis therefore uses an explicit two-week replenishment lead time and 95% cycle-service level (z = 1.645). Demand variability is estimated from the latest 26 complete weeks.

Safety Stock = z x Weekly Demand Standard Deviation x square root of Lead Time

Reorder Point = Average Weekly Demand x Lead Time + Safety Stock

Product	Safety stock	Reorder point	8-week forecast
Regency Cakestand 3 Tier	149	465	1,428
White Hanging Heart T-Light Holder	818	1,915	5,714
Party Bunting	513	1,156	1,691
Assorted Colour Bird Ornament	1,661	3,213	8,594
Chilli Lights	511	924	4,323

The Assorted Colour Bird Ornament receives the largest modeled buffer because it combines the highest recent average demand with the greatest variability. The Regency Cakestand requires the smallest buffer among the selected products.



4. Recommendations and limitations

Recommended actions

1. Refresh forecasts weekly and compare actual demand with the forecast before releasing replenishment orders.
2. Review lead times and service targets by product instead of using a single blanket inventory rule.
3. Build seasonal inventory before the September-November demand peak while monitoring post-holiday overstock risk.
4. Connect future versions to purchase-order, inventory-balance, supplier lead-time and promotion data.
5. Track forecast bias, MAE/WAPE, service level, stockouts and inventory turns on the operating dashboard.

Limitations

The source does not identify lost sales, stockouts, supplier lead times, on-hand inventory, purchase costs, holding costs or promotions. Inventory outputs are therefore scenario recommendations rather than observed company policies. The eight-week forecasts should not be interpreted as guaranteed demand, especially around holidays and promotions.

Portfolio skills demonstrated

Supply chain analytics | Demand forecasting | Inventory optimization | Python | pandas | scikit-learn | Excel dashboarding | Data visualization | Business communication

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